

CoalMind AI

Prototype & working report

Smart India Hackathon 2025 · Problem SIH 26023

AI-Powered Geological, Mining and other Reporting Solution for CMPDI / CIL subsidiaries

Team AGNIVAULT · captured 26 September 2026

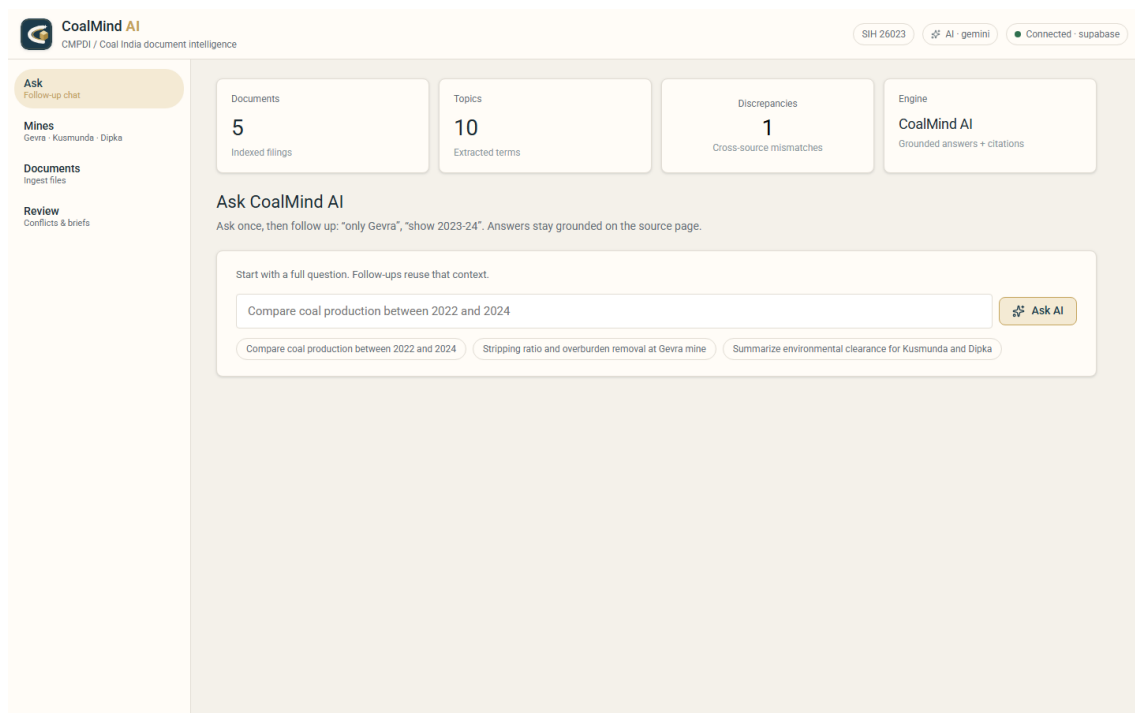


Figure 1. Working console — Ask home, Gemini connected, 5 indexed filings.

1. Problem

CMPDI and Coal India subsidiaries produce large volumes of PDFs, mine registers, and spreadsheets. Officers still re-type figures, lose the page a number came from, and miss when two filings disagree on the same mine and year. SIH 26023 asks for an AI reporting copilot that extracts, retrieves, verifies, and reports — not a chatbot that invents tonnes.

2. What the prototype is

Upload → Extract → Store → Ask AI → Cite the page → Flag conflicts → Download a brief. The stack runs on a normal laptop (no CUDA, torch, or FAISS). Gemini is used only as a language layer on top of indexed archive rows. In this capture: 5 documents, 10 topics, 1 discrepancy, store = Supabase, AI = Gemini.

3. How it works

PDF / XLSX / CSV → parser (pypdf / pandas) → text chunks + mine/year/parameter metrics → Supabase or SQLite. SQL intent charts structured metrics. RAG intent searches hashed 384-d chunks. A conflict scan flags the same mine + year + metric in two files when the gap exceeds 5%. Gemini then writes English from those facts only. If the archive

has no row, the answer says so. Short follow-ups such as “Only Gevra” are expanded onto the previous question.

4. Ask — working query

Question: *Compare coal production between 2022 and 2024.* The thread returns a Gemini briefing with source file and page, a bar chart from indexed metrics, an officer briefing card (three bullets, table, citation, Download as note), and a source list with View.

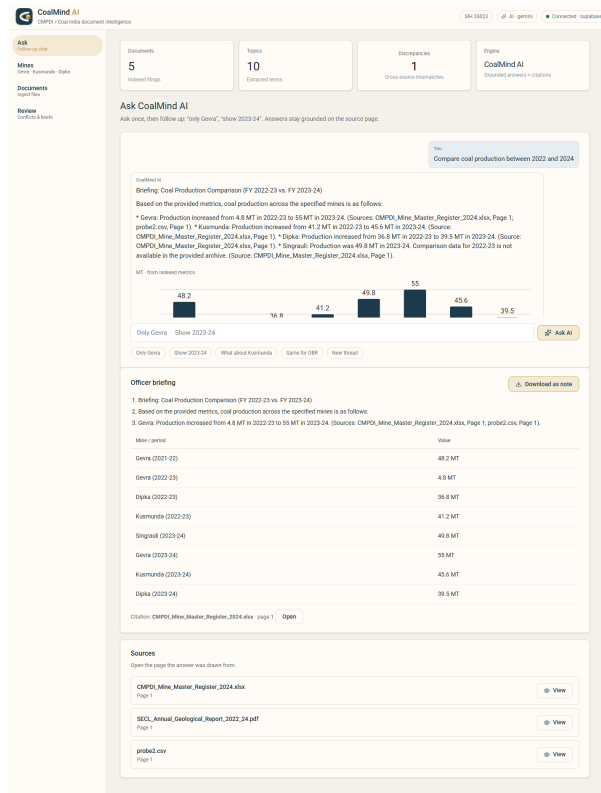


Figure 2. Live Ask result — cited figures, chart, officer note, sources.

5. Mine dashboard

One jury screenshot: Gevra, Kusunda, and Dipka with latest production, overburden, stripping ratio, and last source file.

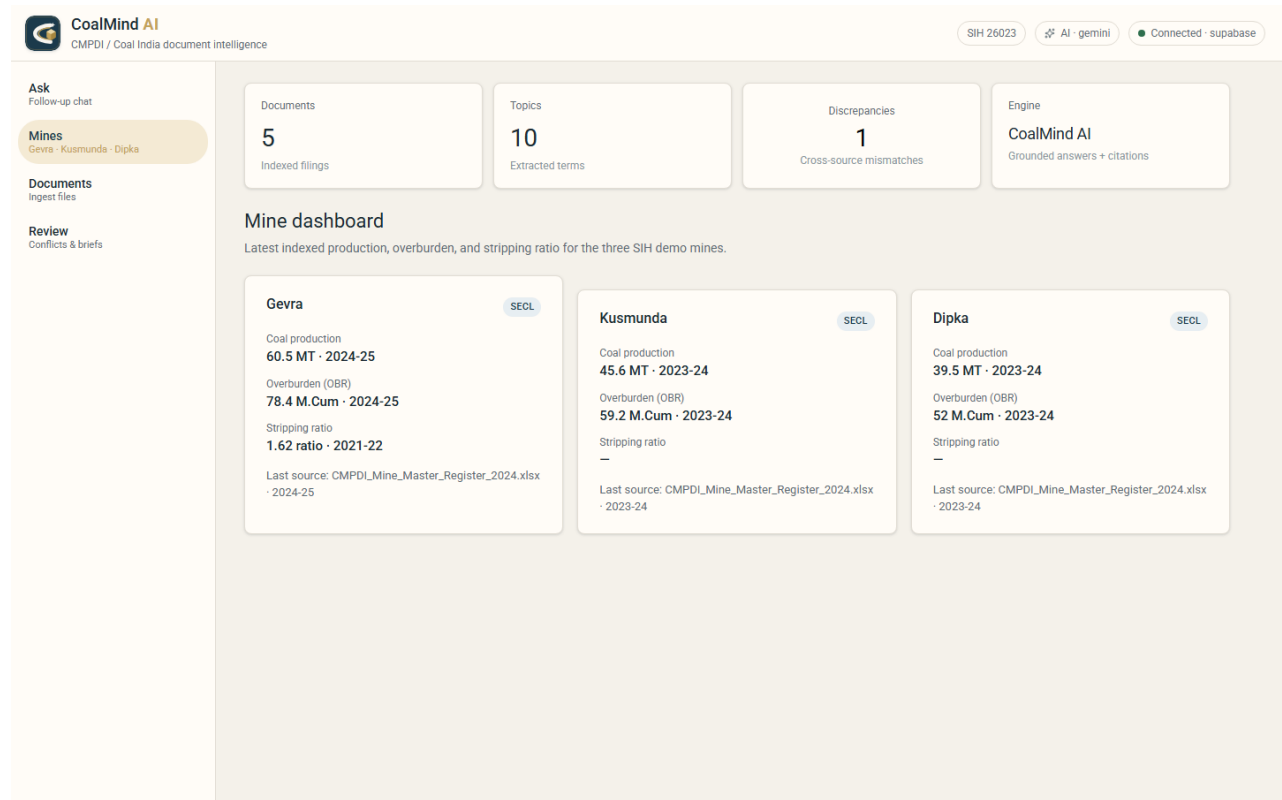


Figure 3. Mine dashboard (SECL demo mines).

Mine	Production	OBR	Last source
Gevra	60.5 MT · 2024-25	78.4 M.Cum	Mine master register
Kusmunda	45.6 MT · 2023-24	59.2 M.Cum	Mine master register
Dipka	39.5 MT · 2023-24	52.0 M.Cum	Mine master register

6. Documents and SIH sample pack

Officers ingest PDF, Excel, or CSV (25 MB). Load SIH sample (SECL 2022–24) seeds SECL_Annual_Geological_Report_2022_24.pdf and CMPDI_Mine_Master_Register_2024.xlsx so the stage demo does not depend on a USB stick. Scanned files without OCR are marked ocr_unavailable — text is not invented.

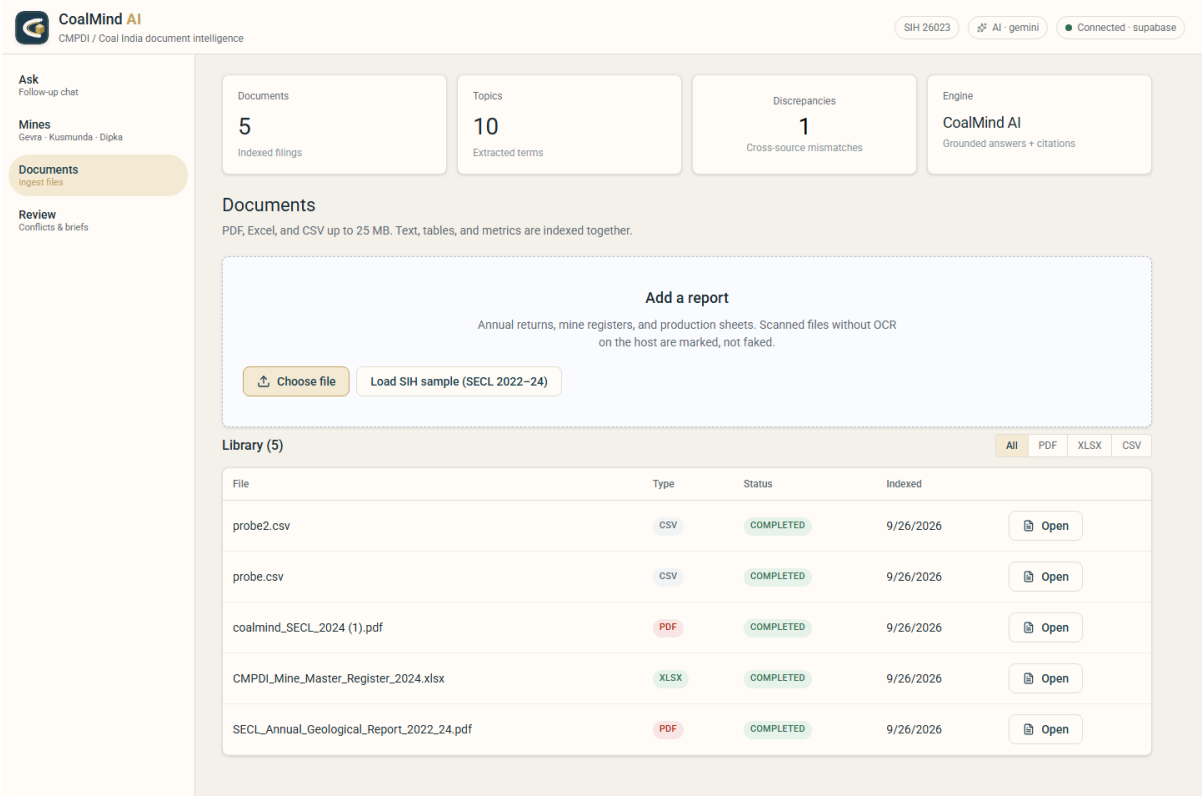


Figure 4. Document library and one-click SIH sample pack.

7. Review — conflicts and PDF brief

Gevra · SECL · 2021-22 · Overburden Removal: 68.4 in the annual PDF versus 62.1 in the mine register (tolerance 5%). Officer actions: Accept, Flag for field check, Ignore. This capture shows FIELD_CHECK. The right panel builds a one-page executive PDF (ReportLab) for a subsidiary and financial year.

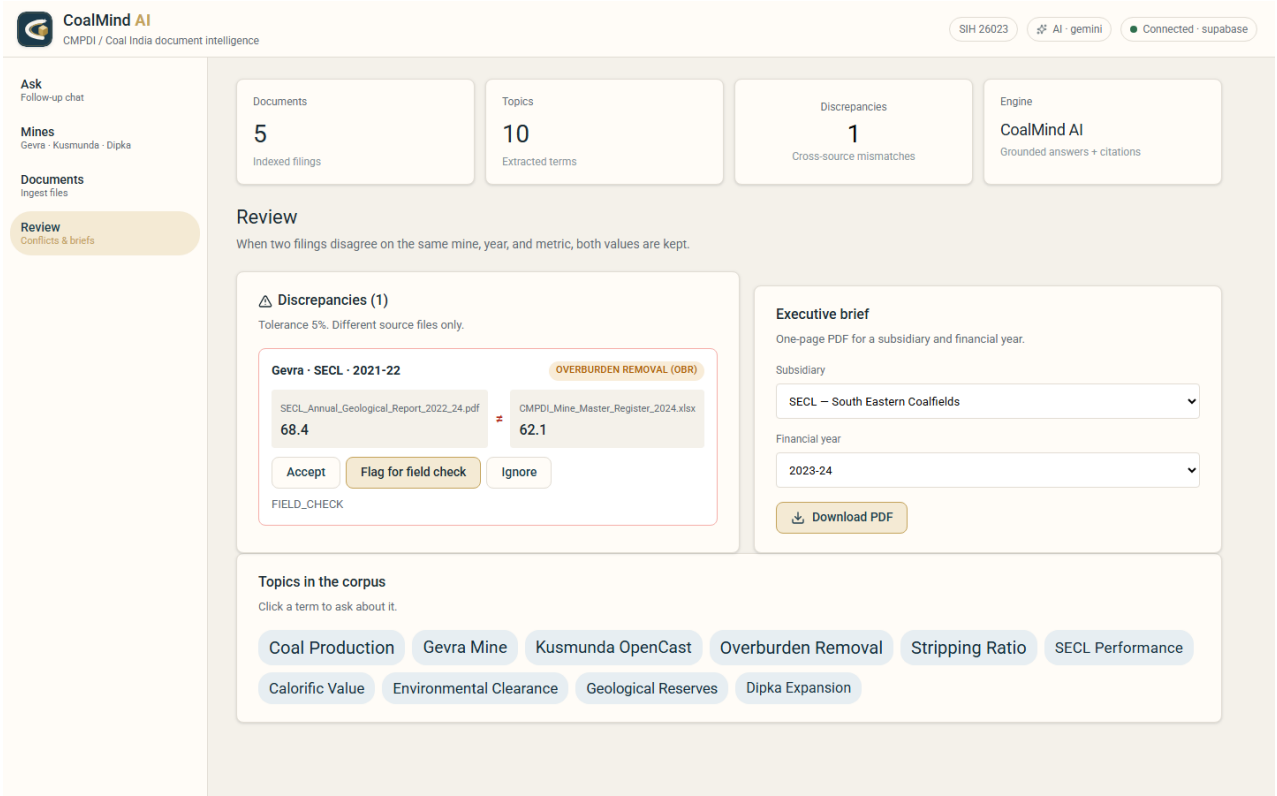


Figure 5. Conflict control room and executive PDF download.

8. Five-minute jury script

- 1. Open <http://127.0.0.1:5173> — show AI · gemini and document count.
- 2. Documents → Load SIH sample if the library is empty.
- 3. Ask: Compare coal production between 2022 and 2024.
- 4. Show the chart and Download as note.
- 5. Follow-up chip: Only Gevra.
- 6. Mines — three tiles.
- 7. Review — Gevra OBR 68.4 ≠ 62.1 → Flag for field check.
- 8. Download PDF for SECL 2023-24.

9. Stack and limits

Layer	Choice
UI	React + Vite :5173
API	FastAPI :8000
Extract	pypdf, pandas, regex
Vectors	Hashed / SVD 384-d (no torch)
LLM	Gemini 3.8 Flash via httpx (optional)
Data	Supabase Postgres or SQLite
Hosted UI	GitHub Pages (read-only Ask / library)

Limits (deliberate): no GPU OCR; embeddings are not a large model; GitHub Pages cannot ingest files; Gemini is optional; officer roles are not full IAM. The product stays up on an 8 GB laptop.

10. Conclusion

CoalMind AI is a working CMPDI-style console: live Gemini answers, cited metrics, a three-mine dashboard, a one-click SECL sample pack, and a conflict that must be accepted or sent for field check. That is the SIH 26023 story — reports you can audit, not free-form generation.

Local UI: <http://127.0.0.1:5173> Hosted: <https://2410305056-tech.github.io/coalmind-ai/> Repo: <https://github.com/2410305056-tech/coalmind-ai>